Predicting Credit Card Default: A Machine Learning Approach for InclusiFinance

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1. Introduction

InclusiFinance faces the dual challenge of identifying high-risk credit card holders while maintaining its commitment to financial inclusion. With approximately 22% of cardholders defaulting on payments, misclassification carries substantial costs in both directions — undetected defaults result in direct financial losses, while incorrectly flagging creditworthy customers restricts their financial access and damages relationships. This analysis developed and evaluated three machine learning models on historical payment data from 30,000 Taiwanese cardholders to deliver an interpretable, accurate, and operationally efficient risk assessment tool that balances risk mitigation with equitable lending.

2. Data & Methodology

The dataset, sourced from the UCI Machine Learning Repository, contains 30,000 credit card accounts from Taiwan, including six months of repayment history (April–September 2005), bill statements, payment amounts, demographic variables, and credit limit. The target variable indicates default in October 2005, with a 77.9% / 22.1% non-default / default split. Data was divided 70/15/15 into stratified training, validation, and test sets. Undocumented categorical codes were recoded to 'other,' and all features were standardized using StandardScaler.

Three models were evaluated, each addressing class imbalance differently. KNN used SMOTE to balance the training set to 16,355 examples per class, operating on a reduced 13-feature set (replacing individual bill and payment amount columns with their means) to mitigate the curse of dimensionality; k=30 was selected by validation AUC-ROC. The Random Forest used scikit-learn's built-in class_weight='balanced' parameter on all 23 features, with grid search over 108 hyperparameter combinations yielding 300 trees, max depth 10, min samples per leaf 4, and sqrt feature selection (val AUC-ROC = 0.7767). The Neural Network used two hidden layers (64→32 units, ReLU, 30% dropout) with computed class weights (0.64 / 2.26) and early stopping after 28 epochs to prevent overfitting.

3. Results & Analysis

On the held-out test set of 4,500 accounts, the tuned Random Forest achieved the strongest overall performance: 78.9% accuracy, AUC-ROC of 0.7771, default recall of 58.1%, precision of 52.1%, and a default F1 of 0.5493. The Neural Network was competitive (AUC-ROC 0.7703, F1 0.5317) but offered no meaningful advantage over the simpler ensemble. KNN with SMOTE achieved the highest default recall at 63.6%, but its 40.3% precision means roughly 60% of flagged accounts were false positives — an unacceptable rate for a financial inclusion mandate.

The ROC curves in Figure 1 visually confirm these findings. The Random Forest and Neural Network track closely together, both substantially outperforming KNN — particularly at low false-positive rates where precision is most critical for protecting creditworthy customers from erroneous denials.

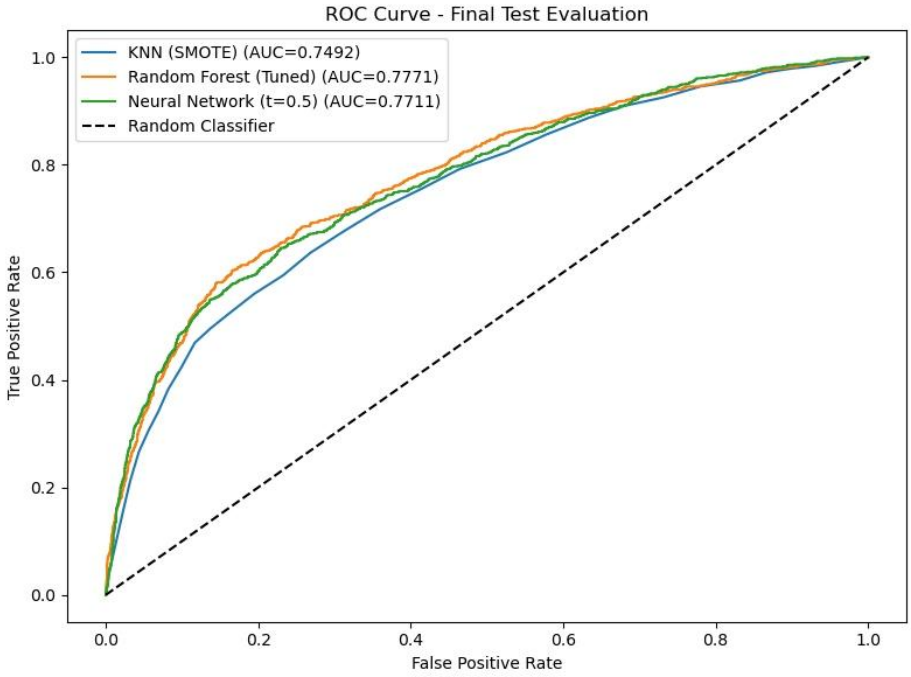


Figure 1. ROC curves on the final test set. Random Forest (AUC = 0.7771) and Neural Network (AUC = 0.7711) demonstrate nearly identical discriminatory power, substantially outperforming KNN (AUC = 0.7492).

The Random Forest's feature importance analysis identified PAY_0 (most recent repayment status) as the dominant predictor by a wide margin, followed by LIMIT_BAL and PAY_2 through PAY_4. Crucially, demographic features — gender, education, marital status — showed minimal importance, confirming that the model evaluates customers primarily on demonstrated financial behavior rather than potentially biased demographic characteristics. This directly supports InclusiFinance's equitable lending mission.

4. Model Comparison & Recommendation

The Random Forest's interpretability is a decisive advantage for production deployment. When an account is flagged as high-risk, credit analysts can trace the decision directly to specific features — for example: 'This account was flagged due to a two-month payment delay in September (PAY_0 = 2), combined with recurring delays in the preceding months.' This level of specificity enables compliance with regulatory requirements for explainable automated credit decisions and supports transparent customer communication. In contrast, the Neural Network requires advanced tools like SHAP values to generate comparable explanations, adding complexity and overhead.

Training efficiency further distinguishes the models. The Random Forest trains in approximately 2 seconds on the 21,000-row dataset, making monthly or weekly retraining feasible on standard hardware with no specialized infrastructure. The Neural Network required 45–60 seconds per fit and substantially longer total development time across architecture selection, learning rate tuning, and threshold optimization — without delivering a meaningful performance gain. KNN, while fast to fit, scales poorly at prediction time as each inference requires distance calculations against all training points, creating bottlenecks under high-volume batch scoring.

The recommendation is to deploy the tuned Random Forest. It delivers the best overall performance, the highest interpretability, and the lowest operational overhead. The KNN model's 60% false positive rate would

undermine InclusiFinance's inclusion mission. The Neural Network's marginal AUC advantage of 0.0068 does not justify its added complexity, training time, or reduced explainability.

5. Conclusion & Business Implications

The Random Forest model provides InclusiFinance with an accurate, transparent, and operationally practical tool for credit risk management. By identifying 58% of defaults before they occur, the model enables proactive interventions — payment reminders, credit limit adjustments, or restructured plans — that can materially reduce portfolio losses. Its 52% default precision minimizes harm to creditworthy applicants, and its reliance on behavioral rather than demographic signals reinforces equitable lending. Automated risk scoring is estimated to reduce manual underwriting time by 60–80%, freeing analysts to focus on complex borderline cases.

Recommended next steps include integrating the model into the credit decisioning pipeline with a live monitoring dashboard tracking AUC-ROC, recall, and precision monthly. Within three to six months, retraining on InclusiFinance-specific outcome data will improve performance beyond the Taiwanese baseline. Threshold optimization — for example, lowering the classification threshold from 0.5 to 0.45 — can push default recall to approximately 62% for more conservative risk management. Longer-term, implementing SHAP-based individual explanations will further strengthen regulatory compliance and customer transparency.